

How roll, alloc and contiguous fill one sample

Data: ischemia_187_1_raw.mrd2, brightest of 312 series readouts (the averaged prescan is never shifted). 64 switches of 20 samples; the reconstruction reads positions 2..13 of each switch, with the echo expected at 8 (ramp 2 + 12/2).

Measured drift: +0.2021 samples per switch, so the gradient period is really 20.2021 where the stream is cut every 20. The echo walks +12.73 samples over the train. A +5 zero pad first puts switch 0's echo on position 8; the method then pulls every later switch back.

Switch i is displaced by $\text{offset}_i = -\text{slope} * i$ (anchored on switch 0). Output sample j of switch i is therefore read from the fractional position on the contiguous readout

$$\text{source} = i * \text{total} + j - \text{offset}_i = i * \text{total} + j + \text{slope} * i$$

which in general falls between two measured samples. The three methods differ only in which real samples they read around `source` and with what weight:

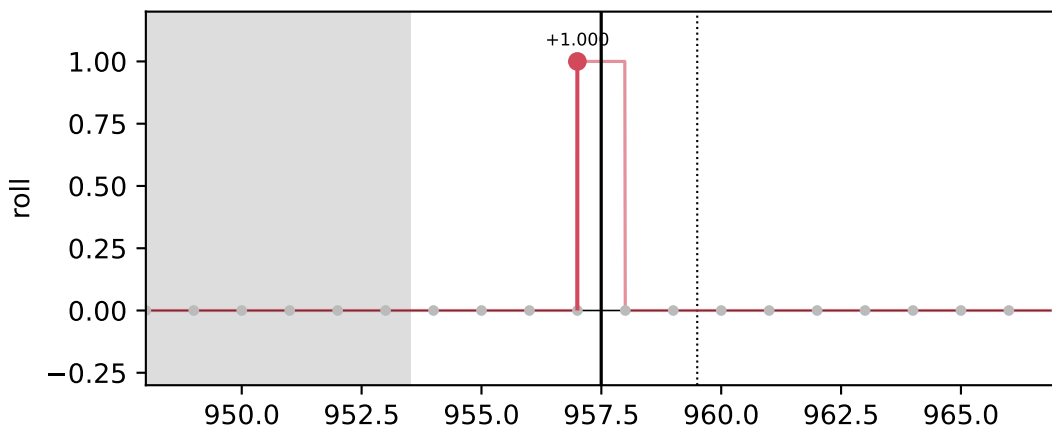
roll	$w = 1$ at $\text{round}(\text{source})$	1 tap
alloc	$w = 1-f$ at $\text{floor}(\text{source})$, f at $\text{floor}+1$	2 taps (linear)
contiguous	$w(x) = \text{sinc}(x) * \text{sinc}(x/8)$, $x = \text{source}-n$	16 taps (Lanczos-8)

f is the fractional part of source. The weights are real, so the same weights are applied to the real and to the imaginary part of the complex signal; the plots show both. All three read along the contiguous readout, so a tap past the end of a switch reads the real neighbouring switch rather than wrapping into its own ramp.

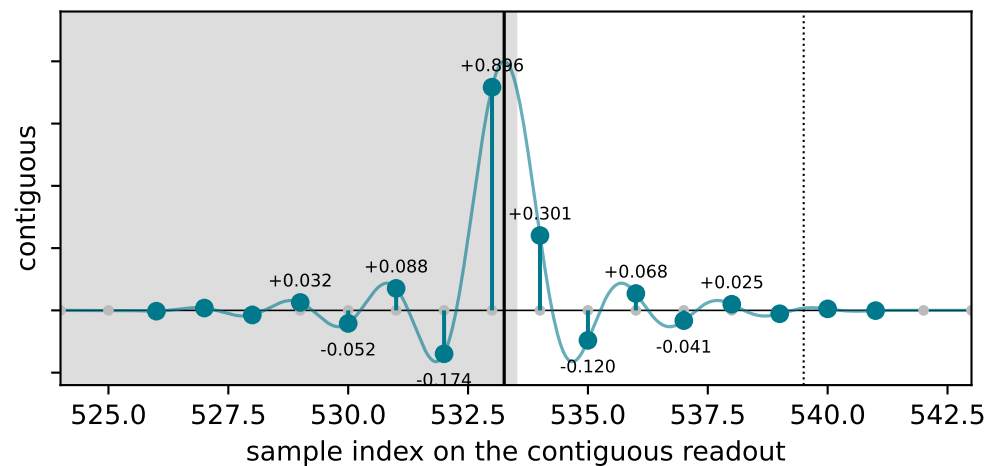
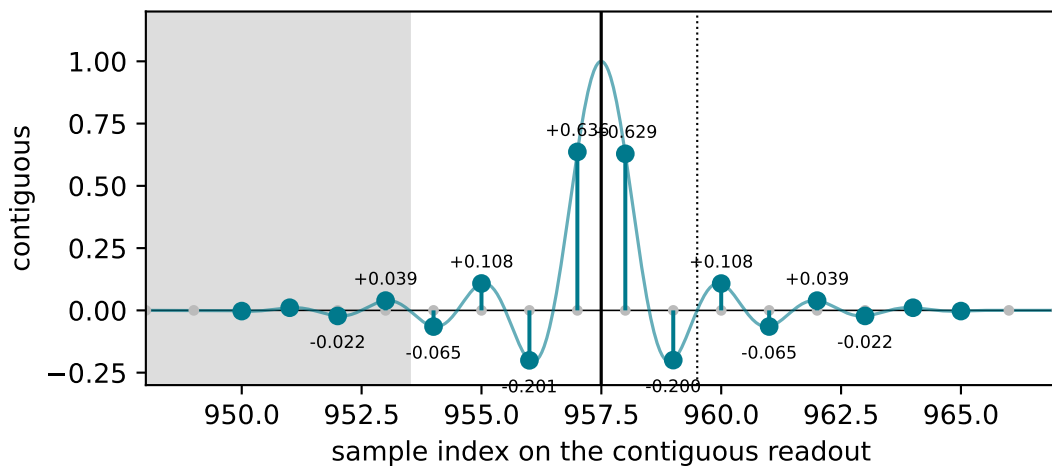
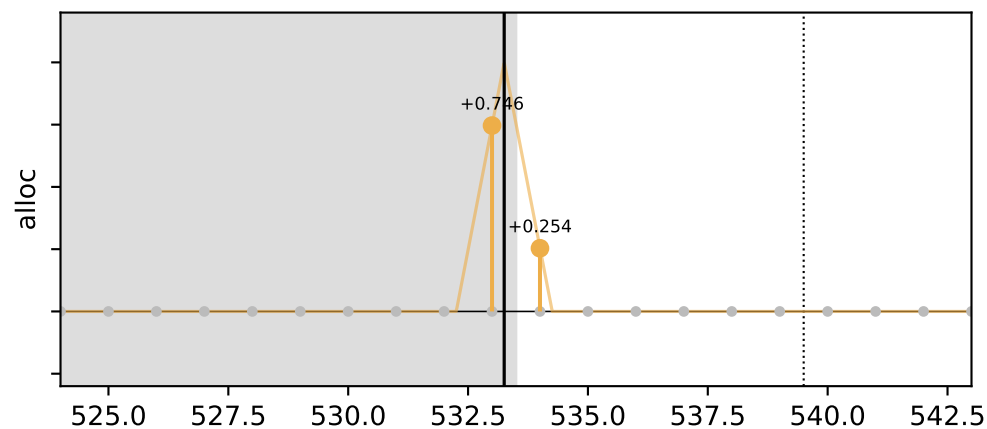
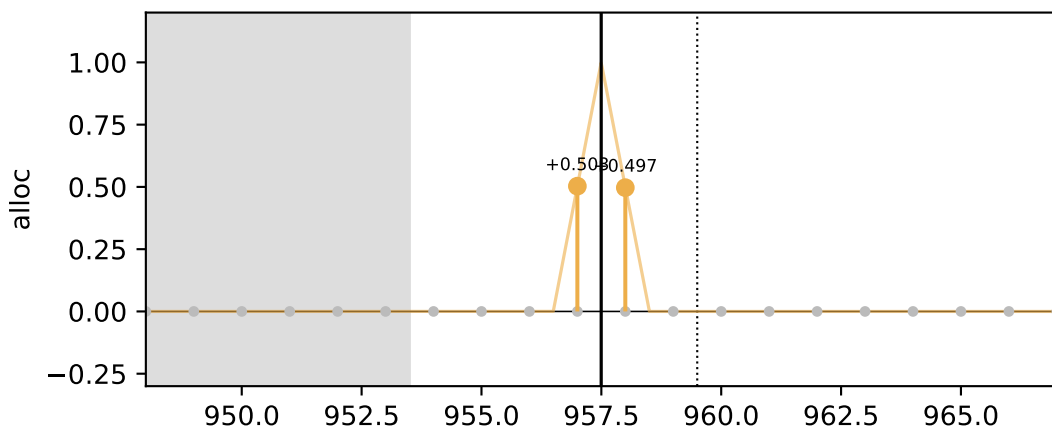
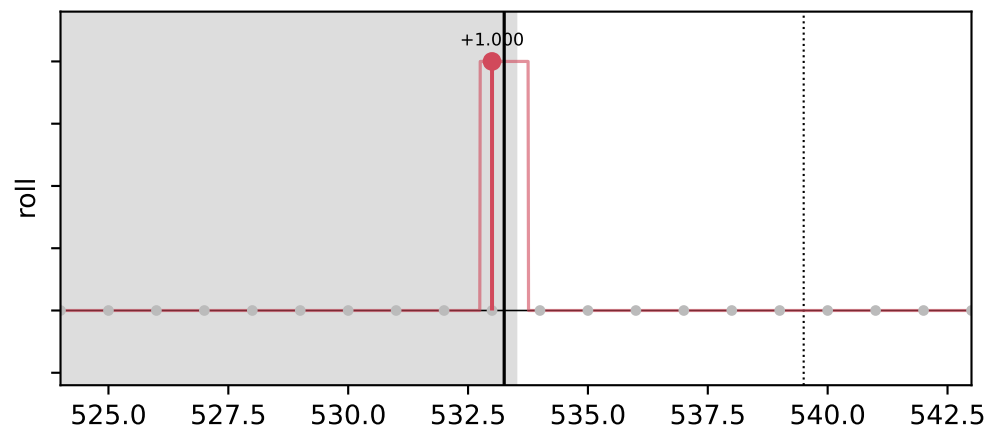
Worked switches: 47 ($f = 0.497$, near one half, where the kernels disagree most) and 26 ($f = 0.254$), both at output position $j = 8$. Every value in this document is checked against `mrd2shift.apply_roll / apply_alloc / apply_contiguous` when it is built.

Weight each method puts on the real samples around the fractional source (black line).
Curve = continuous kernel, stems = the weights actually used; grey band = the source switch's kept window, dotted = switch boundaries

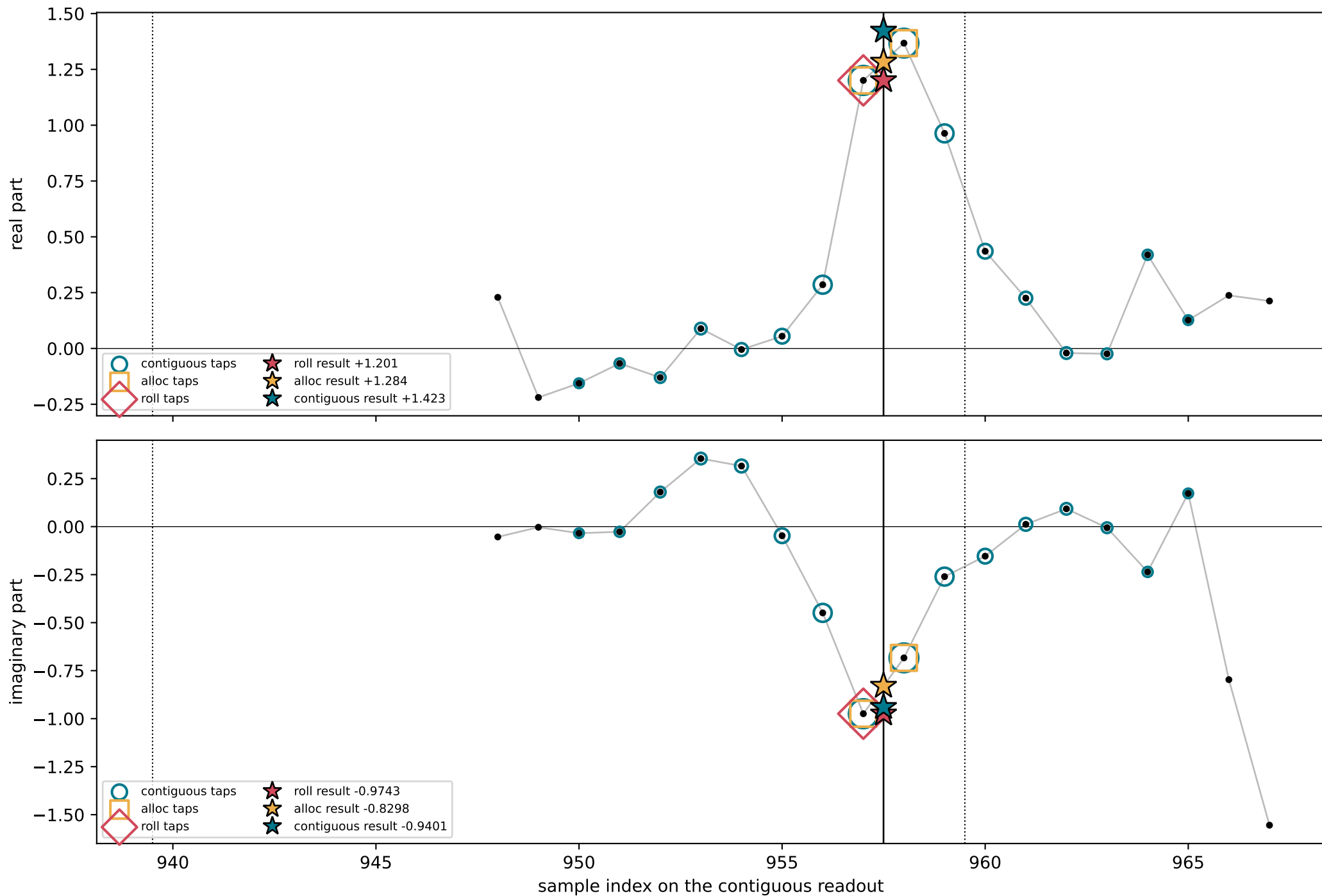
switch 47: source = 957.4970 (f = 0.497)



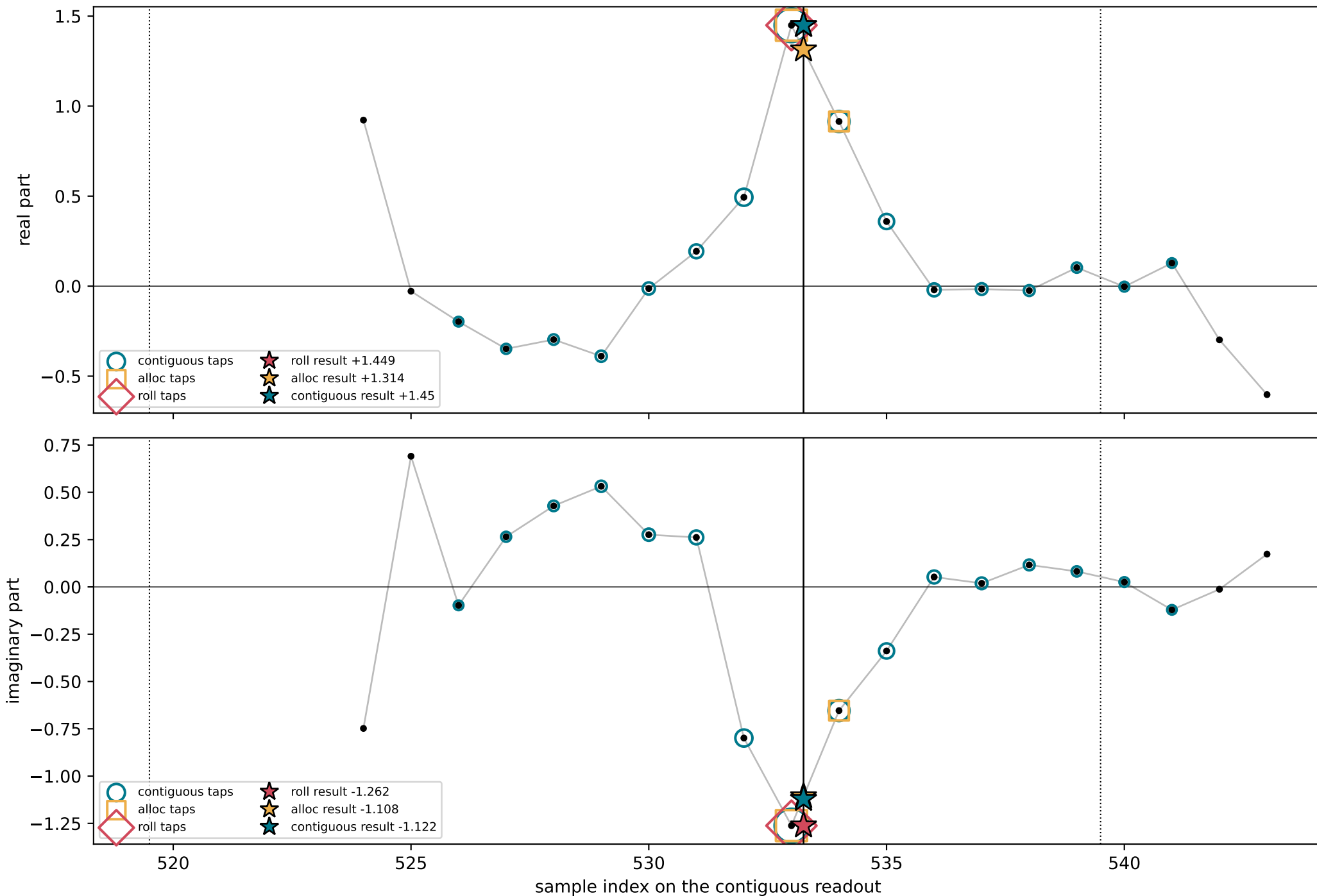
switch 26: source = 533.2537 (f = 0.254)



Switch 47, output position 8: measured samples (black dots), the taps each method reads (rings, size = |weight|), and what each returns at source = 957.4970 (stars). Same real weights on both parts.



Switch 26, output position 8: measured samples (black dots), the taps each method reads (rings, size = |weight|), and what each returns at source = 533.2537 (stars). Same real weights on both parts.



Switch 47: every tap and the weighted sums

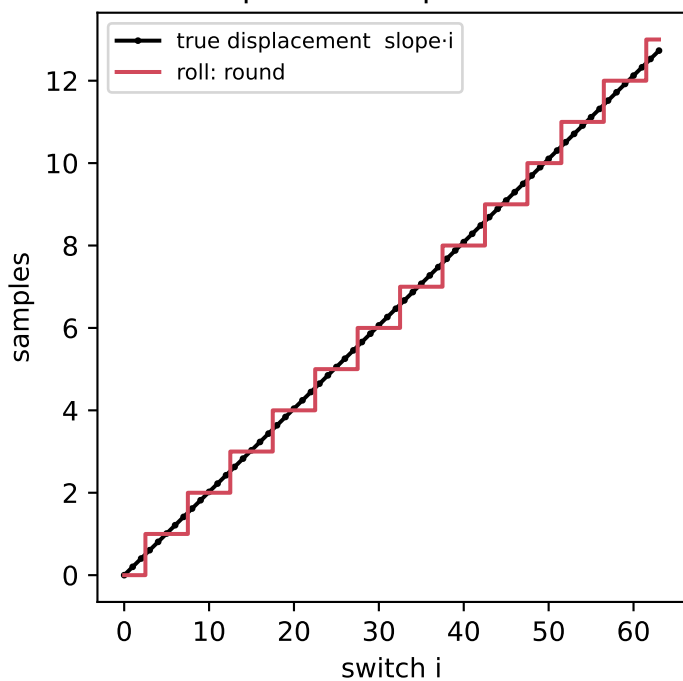
source = 957.4970, $f = 0.4970$. Columns: sample index on the contiguous readout, its position within its switch, the weight roll, alloc and contiguous put on it, and its real and imaginary value. Each sum is $\sum w \cdot (re + i \cdot im)$ and equals the method's output.

Magnitude / phase of each result: roll 1.546 / -39.1° alloc 1.528 / -32.9° contiguous 1.705 / -33.5°

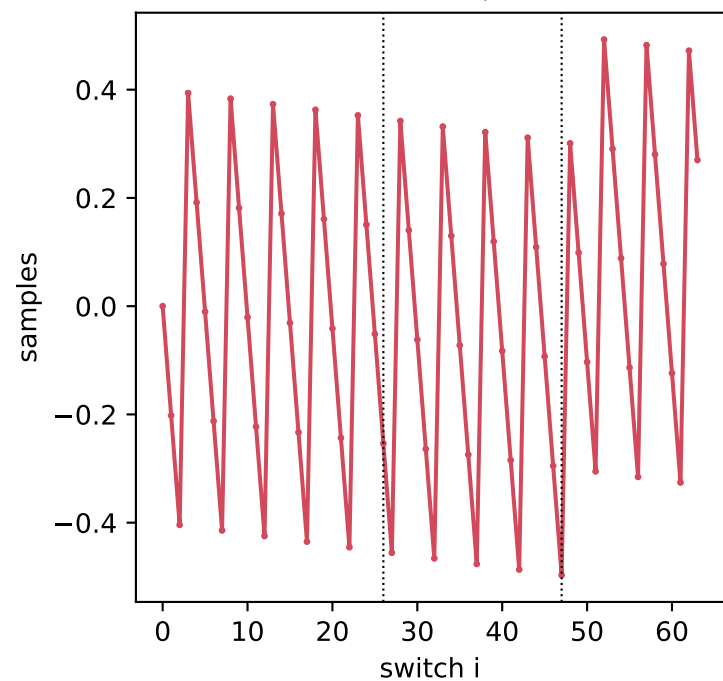
index	pos	roll w	alloc w	contiguous w	real	imag
950	10	0.000	0.000	-0.0028	-0.1559	-0.03424
951	11	0.000	0.000	+0.0107	-0.06708	-0.02696
952	12	0.000	0.000	-0.0223	-0.1303	+0.1799
953	13	0.000	0.000	+0.0393	+0.08887	+0.3544
954	14	0.000	0.000	-0.0650	-0.00479	+0.3157
955	15	0.000	0.000	+0.1080	+0.05484	-0.04737
956	16	0.000	0.000	-0.2006	+0.2855	-0.4492
957	17	1.000	0.503	+0.6364	+1.201	-0.9743
958	18	0.000	0.497	+0.6287	+1.368	-0.6835
959	19	0.000	0.000	-0.1997	+0.9634	-0.2604
960	0	0.000	0.000	+0.1077	+0.4354	-0.1539
961	1	0.000	0.000	-0.0648	+0.2253	+0.01191
962	2	0.000	0.000	+0.0392	-0.02056	+0.09268
963	3	0.000	0.000	-0.0222	-0.02397	-0.005709
964	4	0.000	0.000	+0.0106	+0.4195	-0.2359
965	5	0.000	0.000	-0.0028	+0.1269	+0.173
roll sum					+1.201	-0.9743
alloc sum					+1.284	-0.8298
contiguous sum					+1.423	-0.9401

Across the train: 64 switches, slope +0.2021

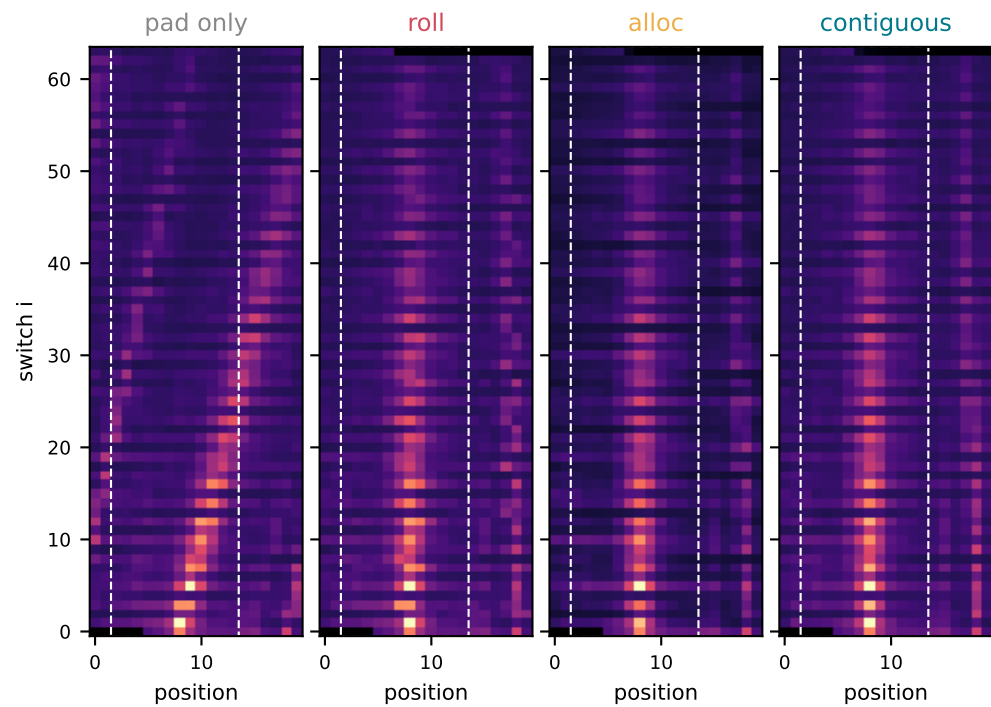
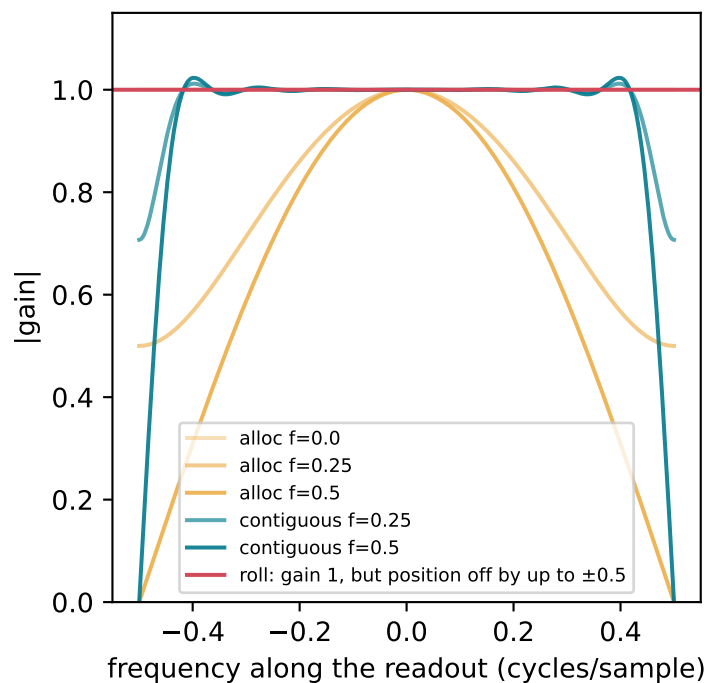
displacement per switch



roll error: true - rounded, a ± 0.5 sawtooth



what each kernel does to the readout's content



What the weights mean

roll reads one measured sample, the nearest one. It is exact only when f is 0; elsewhere it reads a point up to half a sample from where the echo really is. Its gain is 1 at every frequency, so it does not blur, but it leaves a position error that saws between -0.5 and +0.5 along the train. The switch index is the spectral axis, so that per-switch error lands in the spectrum as a periodic phase error.

alloc is a two-tap triangle: $1-f$ on the sample below, f on the sample above. It lands on the right position on average, but a straight line between two samples is a low-pass filter. At $f = 0.5$ it simply averages two neighbours and passes nothing at the Nyquist frequency. The loss changes with f , so it too varies from switch to switch.

contiguous is the band-limited answer: a sinc windowed to 16 taps (Lanczos-8). The weights alternate in sign and fall off with distance, the negative lobes being what restores the high-frequency content the triangle loses. Its gain is flat to near Nyquist at any f , so every switch is moved by the right fraction with the same response. Its taps near the ends of a switch reach into the real neighbouring switch; only at the two physical ends of the readout are they zero.

All three are the same operation, $\sum w_n \cdot x_n$ at the same fractional source, with a different set of weights. That is the whole difference between them.